Chapter 3

Algebraic Expressions

Algebraic formulae are used to solve many algebraic problems. Moreover, many algebraic expressions are presented by resolving them into factors. That is why the problem solved by algebraic formulae and the contents of resolving expressions into factors by making suitable for the students have been presented in this chapter. Moreover, different types of mathematical problems can be solved by resolving into factors with the help of algebraic formulae. In the previous class, algebraic formulae and their related corollaries have been discussed elaborately. In this chapter, those are reiterated and some of their applications are presented through examples. Besides, extension of the formulae of square and cube, resolution into factors using remainder theorem and formation of algebraic formulae and their applications in solving practical problems have been discussed here in detail.

At the end of the chapter, the students will be able to —

- expand the formulae of square and cube by applying algebraic formulae
- explain the remainder theorem and resolve into factors by applying the theorem
- form algebraic formulae for solving real life problems and solve the problems by applying the formulae.

Algebraic Expressions

Meaningful organization of operational signs and numerical letter symbols is called **Algebraic Expressions**. Such as, $2a + 3b - 4c$ is an algebraic expression. In algebraic expression, different types of information are expressed through the letters a, b, c, p, q, r, m, n, x, y, z, ... etc. These alphabet are used to solve different types of problems related to algebraic expressions. In arithmetic, only positive numbers are used, where as, in algebra, both positive and negative

numbers including zero are used. **Algebra is the generalization of arithmetic**.

The numbers used in algebraic expressions are **constants**, their values are fixed.The letter symbols used in algebraic expressions are **variables**, their values are not fixed, they can be of any value.

Algebraic Formulae

Any general rule or resolution expressed by algebraic symbols is called **Algebraic Formula**. In class VII and VIII, algebraic formulae and related corollaries have been discussed. In this chapter, some applications are presented on the basis of that discussion.

Formula 1. $(a+b)^2 = a^2 + 2ab + b^2$

Formula 2. $(a-b)^2 = a^2 - 2ab + b^2$

Remark: It is seen from Formula 1 and Formula 2 that, adding $2ab$ or $-2ab$ to $a^2 + b^2$, we get a perfect square, i.e. we get $(a+b)^2$ or $(a-b)^2$. Substituting $-b$ instead of b in Formula 1 we get Formula 2 : $\{a+(-b)\}^2 = a^2 + 2a(-b) + (-b)^2$ That is, $(a-b)^2 = a^2 - 2ab + b^2$

Corollary 1. $a^2 + b^2 = (a+b)^2 - 2ab$

Corollary 2. $a^2 + b^2 = (a-b)^2 + 2ab$

Corollary 3. $(a+b)^2 = (a-b)^2 + 4ab$

Proof: $(a+b)^2 = a^2 + 2ab + b^2 = a^2 - 2ab + b^2 + 4ab = (a-b)^2 + 4ab$ □

Corollary 4. $(a-b)^2 = (a+b)^2 - 4ab$

Proof: $(a-b)^2 = a^2 - 2ab + b^2 = a^2 + 2ab + b^2 - 4ab = (a+b)^2 - 4ab$ □

Corollary 5. $a^2 + b^2 = \dfrac{(a+b)^2 + (a-b)^2}{2}$

Proof: From Formula 1 and Formula 2,

$$\begin{aligned} a^2 + 2ab + b^2 &= (a+b)^2 \\ a^2 - 2ab + b^2 &= (a-b)^2 \\ \hline \text{Adding, } 2a^2 + 2b^2 &= (a+b)^2 + (a-b)^2 \\ \text{or, } 2(a^2+b^2) &= (a+b)^2 + (a-b)^2 \\ \text{Hence, } (a^2+b^2) &= \frac{(a+b)^2 + (a-b)^2}{2} \end{aligned}$$ □

Corollary 6. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$

Proof: From Formula 1 and Formula 2,

$$\begin{aligned} a^2 + 2ab + b^2 &= (a+b)^2 \\ a^2 - 2ab + b^2 &= (a-b)^2 \\ \hline \text{Subtracting, } 4ab &= (a+b)^2 - (a-b)^2 \\ \text{or, } ab &= \frac{(a+b)^2}{4} - \frac{(a-b)^2}{4} \\ \text{hence, } ab &= \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 \end{aligned}$$ □

Remark: By applying the Corollary 6 product of any two quantities can be expressed as the difference of two squares.

Formula 3. $a^2 - b^2 = (a+b)(a-b)$

Therefore, the difference of the squares of two expressions = sum of two expressions × difference of two expressions.

Formula 4. $(x+a)(x+b) = x^2 + (a+b)x + ab$

Therefore, $(x+a)(x+b) = x^2$+ (algebraic sum of a and b) x + (the product of a b)

Extension of formula for square: There are three terms in the expression $a+b+c$. It can be considered the sum of two terms $(a+b)$ and c. Therefore, by applying Formula 1, the square of the expression is,

$$(a+b+c)^2 = \{(a+b)+c\}^2 = (a+b)^2 + 2(a+b)c + c^2$$

$$= a^2 + 2ab + b^2 + 2ac + 2bc + c^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$$

Formula 5. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$

Corollary 7. $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ac)$

Corollary 8. $2(ab+bc+ac) = (a+b+c)^2 - (a^2+b^2+c^2)$

Note: Applying Formula 5, we get,

1) $(a+b-c)^2 = \{a+b+(-c)\}^2$
$= a^2+b^2+(-c)^2+2ab+2b(-c)+2a(-c)$
$= a^2+b^2+c^2+2ab-2bc-2ac$

2) $(a-b+c)^2 = \{a+(-b)+c\}^2$
$= a^2+(-b)^2+c^2+2a(-b)+2(-b)c+2ac$
$= a^2+b^2+c^2-2ab-2bc+2ac$

3) $(a-b-c)^2 = \{a+(-b)+(-c)\}^2$
$= a^2+(-b)^2+(-c)^2+2a(-b)+2(-b)(-c)+2a(-c)$
$= a^2+b^2+c^2-2ab+2bc-2ac$

Example 1. What is the square of $(4x+5y)$?

Solution: $(4x+5y)^2 = (4x)^2+2\times(4x)\times(5y)+(5y)^2 = 16x^2+40xy+25y^2$

Example 2. What is the square of $(3a-7b)$?

Solution: $(3a-7b)^2 = (3a)^2-2\times(3a)\times(7b)+(7b)^2 = 9a^2-42ab+49b^2$

Example 3. Find the square of 996 by applying the formula of square.

Solution: $(996)^2 = (1000-4)^2 = (1000)^2-2\times1000\times4+4^2$

$= 1000000-8000+16 = 1000016-8000 = 992016$

Example 4. What is the square of $a+b+c+d$?

Solution: $(a+b+c+d)^2 = \{(a+b)+(c+d)\}^2$

$= (a+b)^2+2(a+b)(c+d)+(c+d)^2$

$= a^2+2ab+b^2+2(ac+ad+bc+bd)+c^2+2cd+d^2$

$= a^2+b^2+c^2+d^2+2ab+2ac+2ad+2bc+2bd+2cd$

Work: Find the square with the help of the formulae:

1) $3xy+2ax$ 2) $4x-3y$ 3) $x-5y+2z$

Example 5. Simplify: $(5x+7y+3z)^2+2(7x-7y-3z)(5x+7y+3z)+(7x-7y-3z)^2$

Solution: Let, $5x + 7y + 3z = a$ and $7x - 7y - 3z = b$

$\therefore$ Given expression $= a^2 + 2 \cdot b \cdot a + b^2 = a^2 + 2ab + b^2$

$= (a + b)^2$

$= \{(5x + 7y + 3z) + (7x - 7y - 3z)\}^2$[Substituting the value of a and b

$= (5x + 7y + 3z + 7x - 7y - 3z)^2$

$= (12x)^2 = 144x^2$

Example 6. If $x - y = 2$ and $xy = 24$, what is the value of $x + y$?

Solution: $(x + y)^2 = (x - y)^2 + 4xy = (2)^2 + 4 \times 24 = 4 + 96 = 100$

$\therefore x + y = \pm\sqrt{100} = \pm 10$

Example 7. If $a^4 + a^2b^2 + b^4 = 3$ and $a^2 + ab + b^2 = 3$, what is the value of $a^2 + b^2$?

Solution: $a^4 + a^2b^2 + b^4$

$= (a^2)^2 + 2a^2b^2 + (b^2)^2 - a^2b^2$

$= (a^2 + b^2)^2 - (ab)^2$

$= (a^2 + b^2 + ab)(a^2 + b^2 - ab)$

$= (a^2 + ab + b^2)(a^2 - ab + b^2)$

$\therefore 3 = 3(a^2 - ab + b^2)$ [Substituting the values]

or, $a^2 - ab + b^2 = \dfrac{3}{3} = 1$

Now adding, $a^2 + ab + b^2 = 3$ and $a^2 - ab + b^2 = 1$

we get, $2(a^2 + b^2) = 4$

or, $a^2 + b^2 = \dfrac{4}{2} = 2$

$\therefore a^2 + b^2 = 2$

Example 8. Prove that, $(a + b)^4 - (a - b)^4 = 8ab(a^2 + b^2)$

Solution: $(a + b)^4 - (a - b)^4$

$= \{(a + b)^2\}^2 - \{(a - b)^2\}^2$

$= \{(a + b)^2 + (a - b)^2\}\{(a + b)^2 - (a - b)^2\}$

$= 2(a^2 + b^2) \times 4ab$ [Applying Corollary 5 and Corollary 6]

$= 8ab(a^2 + b^2)$

$\therefore (a+b)^4 - (a-b)^4 = 8ab(a^2+b^2)$

Example 9. If $a + b + c = 15$ and $a^2 + b^2 + c^2 = 83$, what is the value of $ab + bc + ac$?

Solution: First method:

$2(ab+bc+ac) = (a+b+c)^2 - (a^2+b^2+c^2) = (15)^2 - 83 = 225 - 83 = 142$

$\therefore ab + bc + ac = \dfrac{142}{2} = 71$

Alternative method:

$(a+b+c)^2 = (a^2+b^2+c^2) + 2(ab+bc+ac)$

or, $(15)^2 = 83 + 2(ab+bc+ac)$

or, $225 - 83 = 2(ab+bc+ac)$

or, $2(ab+bc+ac) = 142$

$\therefore ab + bc + ac = \dfrac{142}{2} = 71$

Example 10. If $a + b + c = 2$ and $ab + bc + ac = 1$, what is the value of $(a+b)^2 + (b+c)^2 + (c+a)^2$?

Solution: $(a+b)^2 + (b+c)^2 + (c+a)^2$

$= a^2 + 2ab + b^2 + b^2 + 2bc + c^2 + c^2 + 2ca + a^2$

$= (a^2 + b^2 + c^2 + 2ab + 2bc + 2ca) + (a^2 + b^2 + c^2)$

$= (a+b+c)^2 + (a+b+c)^2 - 2(ab+bc+ca)$

$= (2)^2 + (2)^2 - 2 \times 1 = 4 + 4 - 2 = 8 - 2 = 6$

Example 11. Express $(2x + 3y)(4x - 5y)$ as the difference of two squares.

Solution: Let, $2x + 3y = a$ and $4x - 5y = b$

$\therefore$ Given expression $ab = \left(\dfrac{a+b}{2}\right)^2 - \left(\dfrac{a-b}{2}\right)^2$

$= \left(\dfrac{2x+3y+4x-5y}{2}\right)^2 - \left(\dfrac{2x+3y-4x+5y}{2}\right)^2$ [Substituting the values of a and b]

$$= \left(\frac{6x-2y}{2}\right)^2 - \left(\frac{8y-2x}{2}\right)^2$$

$$= \left\{\frac{2(3x-y)}{2}\right\}^2 - \left\{\frac{2(4y-x)}{2}\right\}^2$$

$$= (3x-y)^2 - (4y-x)^2$$

$\therefore (2x+3y)(4x-5y) = (3x-y)^2 - (4y-x)^2$

Work:

1) Simplify: $(4x+3y)^2 + 2(4x+3y)(4x-3y) + (4x-3y)^2$

2) If $x+y+z = 12$ and $x^2+y^2+z^2 = 50$, find the value of $(x-y)^2 + (y-z)^2 + (z-x)^2$.

Exercise 3.1

1. Find the square with the help of the formulae:
   1) $2a+3b$ 2) $x^2 + \frac{2}{y^2}$ 3) $4y-5x$
   4) $5x^2-y$ 5) $3b-5c-2a$ 6) $ax-by-cz$
   7) $2a+3x-2y-5z$ 8) 1007
2. Simplify:
   1) $(7p+3q-5r)^2 - 2(7p+3q-5r)(8p-4q-5r) + (8p-4q-5r)^2$
   2) $(2m+3n-p)^2 + (2m-3n+p)^2 - 2(2m+3n-p)(2m-3n+p)$
   3) $6.35 \times 6.35 + 2 \times 6.35 \times 3.65 + 3.65 \times 3.65$
   4) $\dfrac{2345 \times 2345 - 759 \times 759}{2345 - 759}$
3. If $a-b = 4$ and $ab = 60$, what is the value of $a+b$?
4. If $a+b = 9m$ and $ab = 18m^2$, what is the value of $a-b$?
5. If $x - \frac{1}{x} = 4$, Prove that, $x^4 + \frac{1}{x^4} = 322$
6. If $2x + \frac{2}{x} = 3$, what is the value of $x^2 + \frac{1}{x^2}$?
7. If $a + \frac{1}{a} = 2$, Show that, $a^2 + \frac{1}{a^2} = a^4 + \frac{1}{a^4}$

8. If $a+b=\sqrt{7}$ and $a-b=\sqrt{5}$, Prove that, $8ab(a^2+b^2)=24$
9. If $a+b+c=9$ and $ab+bc+ca=31$, what is the value of $a^2+b^2+c^2$?
10. If $a^2+b^2+c^2=9$ and $ab+bc+ca=8$, what is the value of $(a+b+c)^2$?
11. If $a+b+c=6$ and $a^2+b^2+c^2=14$, what is the value of $(a-b)^2+(b-c)^2+(c-a)^2$?
12. If $x=3$, $y=4$ and $z=5$, what is the value of $9x^2+16y^2+4z^2-24xy-16yz+12zx$?
13. Express $(a+2b)(3a+2c)$ as the difference of two squares.
14. Express $x^2+10x+24$ as the difference of two squares.
15. If $a^4+a^2b^2+b^4=8$ and $a^2+ab+b^2=4$ find the value of, 1) a^2+b^2, 2) ab

Formulae of Cubes

Formula 6. $(a+b)^3=a^3+3a^2b+3ab^2+b^3=a^3+b^3+3ab(a+b)$

Proof: $(a+b)^3=(a+b)(a+b)^2$

$=(a+b)(a^2+2ab+b^2)$

$=a(a^2+2ab+b^2)+b(a^2+2ab+b^2)$

$=a^3+2a^2b+ab^2+a^2b+2ab^2+b^3$

$=a^3+3a^2b+3ab^2+b^3$

$=a^3+b^3+3ab(a+b)$ □

Corollary 9. $a^3+b^3=(a+b)^3-3ab(a+b)$

Formula 7. $(a-b)^3=a^3-3a^2b+3ab^2-b^3=a^3-b^3-3ab(a-b)$

Proof: $(a-b)^3=(a-b)(a-b)^2$

$=(a-b)(a^2-2ab+b^2)$

$=a(a^2-2ab+b^2)-b(a^2-2ab+b^2)$

$=a^3-2a^2b+ab^2-a^2b+2ab^2-b^3$

$=a^3-3a^2b+3ab^2-b^3$

$=a^3-b^3-3ab(a-b)$ □

Note: Substituting $-b$ instead of b in Formula 6 we get Formula 7:

$\{a + (-b)\}^3 = a^3 + (-b)^3 + 3a(-b)\{a + (-b)\}$

That is, $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

Corollary 10. $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

Formula 8. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Proof: $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

$= (a + b)\{(a + b)^2 - 3ab\}$

$= (a + b)(a^2 + 2ab + b^2 - 3ab)$

$= (a + b)(a^2 - ab + b^2)$ □

Formula 9. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Proof: $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

$= (a - b)\{(a - b)^2 + 3ab\}$

$= (a - b)(a^2 - 2ab + b^2 + 3ab)$

$= (a - b)(a^2 + ab + b^2)$ □

Example 12. Find the cube of $2x + 3y$.

Solution: $(2x + 3y)^3$

$= (2x)^3 + 3(2x)^2 \cdot 3y + 3 \cdot 2x(3y)^2 + (3y)^3$

$= 8x^3 + 3 \cdot 4x^2 \cdot 3y + 3 \cdot 2x \cdot 9y^2 + 27y^3$

$= 8x^3 + 36x^2y + 54xy^2 + 27y^3$

Example 13. Find the cube of $2x - y$.

Solution: $(2x - y)^3$

$= (2x)^3 - 3(2x)^2 \cdot y + 3 \cdot 2x \cdot y^2 - y^3$

$= 8x^3 - 3 \cdot 4x^2 \cdot y + 3 \cdot 2x \cdot y^2 - y^3$

$= 8x^3 - 12x^2y + 6xy^2 - y^3$

Work: Find the cube with the help of the formulae.

1) $3x + 2y$ 2) $3x - 4y$ 3) 397

Example 14. If $x = 37$, what is the value of $8x^3 + 72x^2 + 216x + 216$?

Solution: $8x^3 + 72x^2 + 216x + 216$

$= (2x)^3 + 3 \cdot (2x)^2 \cdot 6 + 3 \cdot 2x \cdot (6)^2 + (6)^3$

$= (2x + 6)^3 = (2 \times 37 + 6)^3$ [Substituting the values]

$= (74 + 6)^3 = (80)^3 = 512000$

Example 15. If $x - y = 8$ and $xy = 5$, what is the value of $x^3 - y^3 + 8(x + y)^2$?

Solution: $x^3 - y^3 + 8(x + y)^2$

$= (x - y)^3 + 3xy(x - y) + 8\{(x - y)^2 + 4xy\}$

$= (8)^3 + 3 \times 5 \times 8 + 8(8^2 + 4 \times 5)$ [Substituting the values]

$= 8^3 + 15 \times 8 + 8(8^2 + 4 \times 5)$

$= 8^3 + 15 \times 8 + 8 \times 84$

$= 8(8^2 + 15 + 84) = 8(64 + 15 + 84)$

$= 8 \times 163 = 1304$

Example 16. If $a = \sqrt{3} + \sqrt{2}$, Prove that, $a^3 + \dfrac{1}{a^3} = 18\sqrt{3}$

Solution: Given that, $a = \sqrt{3} + \sqrt{2}$

$\therefore \ \dfrac{1}{a} = \dfrac{1}{\sqrt{3} + \sqrt{2}}$

$= \dfrac{\sqrt{3} - \sqrt{2}}{(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})}$ [multiplying numerator and denominator by $(\sqrt{3} - \sqrt{2})$]

$= \dfrac{\sqrt{3} - \sqrt{2}}{(\sqrt{3})^2 - (\sqrt{2})^2} = \dfrac{\sqrt{3} - \sqrt{2}}{3 - 2} = \sqrt{3} - \sqrt{2}$

$\therefore \ a + \dfrac{1}{a} = (\sqrt{3} + \sqrt{2}) + (\sqrt{3} - \sqrt{2}) = \sqrt{3} + \sqrt{2} + \sqrt{3} - \sqrt{2} = 2\sqrt{3}$

Now, $a^3 + \dfrac{1}{a^3} = \left(a + \dfrac{1}{a}\right)^3 - 3 \cdot a \cdot \dfrac{1}{a}\left(a + \dfrac{1}{a}\right)$

$= (2\sqrt{3})^3 - 3(2\sqrt{3}) \left[\because a + \dfrac{1}{a} = 2\sqrt{3}\right]$

$= 2^3 \cdot (\sqrt{3})^3 - 3 \times 2\sqrt{3} = 8 \cdot 3\sqrt{3} - 6\sqrt{3}$

$= 24\sqrt{3} - 6\sqrt{3} = 18\sqrt{3}$ (Proved)

Example 17. If $x+y=5,\ xy=6$ and $x>y$

1) Find the value of $2(x^2+y^2)$.
2) Find the value of $x^3-y^3-3(x^2+y^2)$.
3) Find the value of x^5+y^5.

Solution:

1) We know, $2(x^2+y^2)=2\{(x+y)^2-2xy\}$

$=2(5^2-2\cdot 6)=2\times 13=26$

$\therefore 2(x^2+y^2)=26$

2) Given that, $x+y=5$ and $xy=6,\ x>y$

$\therefore x-y=\pm\sqrt{(x+y)^2-4xy}$

$=\pm\sqrt{5^2-4\cdot 6}=\pm\sqrt{25-24}=\pm\sqrt{1}=\pm 1$

$x^3-y^3-3(x^2+y^2)$

$=(x-y)^3+3xy(x-y)-\frac{3}{2}\cdot 2(x^2+y^2)$

$=1^3+3\cdot 6\cdot 1-\frac{3}{2}\cdot 26$ or $(-1)^3+3\cdot 6\cdot(-1)-\frac{3}{2}\cdot 26$

$=1+18-39$ or $-1-18-39$

$=-20$ or -58

$\therefore x^3-y^3-3(x^2+y^2)=-20$ or -58

3) $x+y=5$ and $x-y=\pm 1$

Adding, $2x=6,4$ $\quad\therefore x=\frac{6}{2}=3$ or $\frac{4}{2}=2$

Subtracting, $2y=4,\ 6$ $\quad\therefore y=\frac{4}{2}=2$ or $\frac{6}{2}=3$

$\therefore x^5+y^5=3^5+2^5=243+32=275$ [The result will be same even if $x=2$ and $y=3$]

Work:

1) If $x=-2$, what is the value of $27x^3-54x^2+36x-8$?
2) If $a+b=5$ and $ab=6$, find the value of $a^3+b^3+4(a-b)^2$.
3) If $x=\sqrt{5}+\sqrt{3}$, find the value of $x^3+\frac{1}{x^3}$.

Exercise 3.2

1. Find the cube with the help of the formulae:
   1) $2x^2 + 3y^2$ 2) $7m^2 - 2n$ 3) $2a - b - 3c$
2. Simplify:
   1) $(7x + 3b)^3 - (5x + 3b)^3 - 6x(7x + 3b)(5x + 3b)$
   2) $(a + b + c)^3 - (a - b - c)^3 - 6(b + c)\{a^2 - (b + c)^2\}$
   3) $(m + n)^6 - (m - n)^6 - 12mn(m^2 - n^2)^2$
   4) $(x + y)(x^2 - xy + y^2) + (y + z)(y^2 - yz + z^2) + (z + x)(z^2 - zx + x^2)$
   5) $(2x + 3y - 4z)^3 + (2x - 3y + 4z)^3 + 12x\{4x^2 - (3y - 4z)^2\}$
3. If $a - b = 5$ and $ab = 36$, what is the value of $a^3 - b^3$?
4. If $a^3 - b^3 = 513$ and $a - b = 3$, what is the value of ab?
5. If $x = 19$ and $y = -12$, find the value of $8x^3 + 36x^2y + 54xy^2 + 27y^3$.
6. If $a = 15$, what is the value of $8a^3 + 60a^2 + 150a + 130$?
7. If $a + b = m$, $a^2 + b^2 = n$ and $a^3 + b^3 = p^3$, show that, $m^3 + 2p^3 = 3mn$.
8. If $a + b = 3$ and $ab = 2$, find the value of (a) $a^2 - ab + b^2$ and (b) $a^3 + b^3$.
9. If $a - b = 5$ and $ab = 36$, find the value of (a) $a^2 + ab + b^2$ and (b) $a^3 - b^3$.
10. If $m + \frac{1}{m} = a$, find the value of $m^3 + \frac{1}{m^3}$.
11. If $x - \frac{1}{x} = p$, find the value of $x^3 - \frac{1}{x^3}$.
12. If $a - \frac{1}{a} = 1$, show that, $a^3 - \frac{1}{a^3} = 4$.
13. If $a + b + c = 0$, Show that,
    1) $a^3 + b^3 + c^3 = 3abc$
    2) $\frac{(b + c)^2}{3bc} + \frac{(c + a)^2}{3ca} + \frac{(a + b)^2}{3ab} = 1.$

14. If $p - q = r$, show that, $p^3 - q^3 - r^3 = 3pqr$.

15. If $2x - \frac{2}{x} = 3$, show that, $8\left(x^3 - \frac{1}{x^3}\right) = 63$.

16. If $a = \sqrt{6} + \sqrt{5}$, find the value of $\frac{a^6 - 1}{a^3}$.

17. $x - \frac{1}{x} = \sqrt{3}$ where $x \neq 0$

    1) Prove that, $x^2 - \sqrt{3}x = 1$.

    2) Prove that, $23\left(x^2 + \frac{1}{x^2}\right) = 5\left(x^4 + \frac{1}{x^4}\right)$.

    3) Find the value of $x^6 + \frac{1}{x^6}$.

Resolution into Factors

If an expression is equal to the product of two or more expressions, each of the latter expressions is called a **factor** of the former expression. After finding the possible factors of any algebraic expression and then expressing the expression as the product of these factors are called **factorization** or resolution into factors. The algebraic expressions may consist of one or more terms. So, the factors may also contain one or more terms. Some process of resolving expressions into factors :

Common Factor: If any polynomial expression has common factor in every term, at first they are to be found out. For example,

Example 18. $3a^2b + 6ab^2 + 12a^2b^2 = 3ab(a + 2b + 4ab)$

Example 19. $2ab(x - y) + 2bc(x - y) + 3ca(x - y) = (x - y)(2ab + 2bc + 3ca)$

Perfect Square: An expression can be expressed in the form of a perfect square.

Example 20. Resolve into factors: $4x^2 + 12x + 9$

Solution: $4x^2 + 12x + 9 = (2x)^2 + 2 \times 2x \times 3 + (3)^2$

$= (2x + 3)^2 = (2x + 3)(2x + 3)$

Example 21. Resolve into factors: $9x^2 - 30xy + 25y^2$

Solution: $9x^2 - 30xy + 25y^2$

$= (3x)^2 - 2 \times 3x \times 5y + (5y)^2$

$= (3x - 5y)^2 = (3x - 5y)(3x - 5y)$

Difference of two squares: Expressing an expression as the difference of two squares and then applying the formula $a^2 - b^2 = (a + b)(a - b)$.

Example 22. Resolve into factors: $a^2 - 1 + 2b - b^2$.

Solution: $a^2 - 1 + 2b - b^2 = a^2 - (b^2 - 2b + 1)$

$= a^2 - (b - 1)^2 = \{a + (b - 1)\}\{a - (b - 1)\}$

$= (a + b - 1)(a - b + 1)$

Example 23. Resolve into factors: $a^4 + 64b^4$

Solution: $a^4 + 64b^4 = (a^2)^2 + (8b^2)^2$

$= (a^2)^2 + 2 \times a^2 \times 8b^2 + (8b^2)^2 - 16a^2b^2$

$= (a^2 + 8b^2)^2 - (4ab)^2$

$= (a^2 + 8b^2 + 4ab)(a^2 + 8b^2 - 4ab)$

$= (a^2 + 4ab + 8b^2)(a^2 - 4ab + 8b^2)$

Work: Resolve into factors:

1) $abx^2 + acx^3 + adx^4$ 2) $xa^2 - 144xb^2$ 3) $x^2 - 2xy - 4y - 4$

Middle term factorization: Factors can be determined by applying the formula $x^2 + (a + b)x + ab = (x + a)(x + b)$. In this method, a polynomial of the form $x^2 + px + q$ can be factorized, if two integers a and b can be found so that, $a + b = p$ and $ab = q$. For this, two factors of q with their signs are to be taken whose algebraic sum is p.If $q>0$, a and b will be of same sign and if $q< 0$, a and b will be of opposite sign. To be noted p and q may no be integers.

Example 24. Resolve into factors: $x^2 + 12x + 35$

Solution: $x^2 + 12x + 35 = x^2 + (5 + 7)x + 5 \times 7 = (x + 5)(x + 7)$

Example 25. Resolve into factors: $x^2 + x - 20$

Solution: $x^2 + x - 20 = x^2 + (5-4)x + (5)(-4) = (x+5)(x-4)$

Middle Term Break-Up: By middle term break-up method of polynomial of the form of ax^2+bx+c would be $ax^2+bx+c = (rx+p)(sx+q)$ if $ax^2+bx+c = rsx^2 + (rq+sp)x + pq$. That is, $a = rs$, $b = rq + sp$ and $c = pq$. Hence, $ac = rspq = (rq)(sp)$ and $b = rq + sp$. Therefore, to determine factors of the polynomial $ax^2 + bx + c$, ac that is, the product of the coefficient of x^2 and the term free from x are to be expressed into two such factors whose algebraic sum is equal to b, the coefficient of x.

Example 26. Resolve into factors: $3x^2 - x - 14$

Solution: $3x^2 - x - 14 = 3x^2 - 7x + 6x - 14$

$= x(3x-7) + 2(3x-7) = (3x-7)(x+2)$

Work: Resolve into factors:

1) $x^2 + x - 56$ 2) $16x^3 - 46x^2 + 15x$ 3) $12x^2 + 17x + 6$

Perfect Cube Form: Factors can be determined by expressing an expression in the form of perfect cube.

Example 27. Resolve into factors: $8x^3 + 36x^2y + 54xy^2 + 27y^3$

Solution: $8x^3 + 36x^2y + 54xy^2 + 27y^3$

$= (2x)^3 + 3 \times (2x)^2 \times 3y + 3 \times 2x \times (3y)^2 + (3y)^3$

$= (2x+3y)^3 = (2x+3y)(2x+3y)(2x+3y)$

Formulae of addition or subtraction of two cubes: Factors can be determined by applying the formulae $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Example 28. Resolve into factors: 1) $8a^3 + 27b^3$ 2) $a^6 - 64$

Solution:

1) $8a^3 + 27b^3 = (2a)^3 + (3b)^3$

$= (2a+3b)\{(2a)^2 - 2a \times 3b + (3b)^2\}$

$= (2a+3b)(4a^2 - 6ab + 9b^2)$

2) $a^6 - 64 = (a^2)^3 - (4)^3 = (a^2-4)\{(a^2)^2 + a^2 \times 4 + (4)^2\}$

$= (a^2 - 4)(a^4 + 4a^2 + 16)$

but $a^2 - 4 = a^2 - 2^2 = (a + 2)(a - 2)$

and $a^4 + 4a^2 + 16 = (a^2)^2 + (4)^2 + 4a^2$

$= (a^2 + 4)^2 - 2(a^2)(4) + 4a^2$

$= (a^2 + 4)^2 - 4a^2$

$= (a^2 + 4)^2 - (2a)^2$

$= (a^2 + 4 + 2a)(a^2 + 4 - 2a)$

$= (a^2 + 2a + 4)(a^2 - 2a + 4)$

$\therefore a^6 - 64 = (a + 2)(a - 2)(a^2 + 2a + 4)(a^2 - 2a + 4)$

Alternative method: $a^6 - 64 = (a^3)^2 - 8^2$

$= (a^3 + 8)(a^3 - 8)$

$= (a^3 + 2^3)(a^3 - 2^3)$

$= (a + 2)(a^2 - 2a + 4)(a - 2)(a^2 + 2a + 4)$

$= (a + 2)(a - 2)(a^2 + 2a + 4)(a^2 - 2a + 4)$

Work: Resolve into factors:

1) $2x^4 + 16x$ 2) $8 - a^3 + 3a^2b - 3ab^2 + b^3$ 3) $(a + b)^3 + (a - b)^3$

Factors of the expression with fractional coefficients: Factors of the expression with fractional coefficients can be expressed in several ways. For example, $a^3 + \frac{1}{27} = a^3 + \frac{1}{3^3} = \left(a + \frac{1}{3}\right)\left(a^2 - \frac{a}{3} + \frac{1}{9}\right)$

Again, $a^3 + \frac{1}{27} = \frac{1}{27}(27a^3 + 1) = \frac{1}{27}\{(3a)^3 + (1)^3\} = \frac{1}{27}(3a + 1)(9a^2 - 3a + 1)$

In the second solution, the factors involving the variables are with integral coefficients but the two solutions are same.

$\frac{1}{27}(3a + 1)(9a^2 - 3a + 1) = \frac{1}{3}(3a + 1) \times \frac{1}{9}(9a^2 - 3a + 1)$

$= \left(a + \frac{1}{3}\right)\left(a^2 - \frac{a}{3} + \frac{1}{9}\right)$

Example 29. Resolve into factors: $x^3 + 6x^2y + 11xy^2 + 6y^3$

Solution: $x^3 + 6x^2y + 11xy^2 + 6y^3$

$= \{x^3 + 3 \cdot x^2 \cdot 2y + 3 \cdot x \cdot (2y)^2 + (2y)^3\} - xy^2 - 2y^3$

$= (x + 2y)^3 - y^2(x + 2y) = (x + 2y)\{(x + 2y)^2 - y^2\}$

$= (x + 2y)(x + 2y + y)(x + 2y - y)$

$= (x + 2y)(x + 3y)(x + y) = (x + y)(x + 2y)(x + 3y)$

Work: Resolve into factors:

1) $\frac{1}{2}x^2 + \frac{7}{6}x + \frac{1}{3}$ 2) $a^3 + \frac{1}{8}$ 3) $16x^2 - 25y^2 - 8xz + 10yz$

Exercises 3.3

Resolve into factors (1-30):

1) $ab(x - y) - bc(x - y)$
2) $9x^2 + 24x + 16$
3) $a^4 - 27a^2 + 1$
4) $x^4 - 6x^2y^2 + y^4$
5) $(a^2 - b^2)(x^2 - y^2) + 4abxy$
6) $4a^2 - 12ab + 9b^2 - 4c^2$
7) $a^2 + 6a + 8 - y^2 + 2y$
8) $16x^2 - 25y^2 - 8xz + 10yz$
9) $x^2 + 13x + 36$
10) $x^4 + x^2 - 20$
11) $a^2 - 30a + 216$
12) $a^8 - a^4 - 2$
13) $x^2 - 37x - 650$
14) $9x^2y^2 - 5xy^2 - 14y^2$
15) $4x^4 - 27x^2 - 81$
16) $ax^2 + (a^2 + 1)x + a$
17) $3(a^2 + 2a)^2 - 22(a^2 + 2a) + 40$
18) $(a - 1)x^2 + a^2xy + (a + 1)y^2$
19) $x^3 + 3x^2 + 3x + 2$
20) $a^3 - 6a^2 + 12a - 9$
21) $a^3 - 9b^3 + (a + b)^3$
22) $8x^3 + 12x^2 + 6x - 63$
23) $8a^3 + \frac{b^3}{27}$
24) $\frac{a^6}{27} - b^6$
25) $4a^2 + \frac{1}{4a^2} - 2 + 4a - \frac{1}{a}$
26) $(3a + 1)^3 - (2a - 3)^3$
27) $(x + 2)(x + 3)(x + 4)(x + 5) - 48$
28) $(x - 1)(x - 3)(x - 5)(x - 7) - 65$
29) $2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4$
30) $14(x + z)^2 - 29(x + z)(x + 1) - 15(x + 1)^2$
31) Show that, $(x + 1)(x + 2)(3x - 1)(3x - 4) = (3x^2 + 2x - 1)(3x^2 + 2x - 8)$

Remainder Theorem

In the following example, if $6x^2 - 7x + 5$ is divided by $x - 1$ then what is quotient and remainder?

$$\begin{array}{rlllll} x-1) & 6x^2 & -7x & +5 & (6x-1 \\ & 6x^2 & -6x & & \\ \hline & & -x & +5 & \\ & & -x & +1 & \\ \hline & & & 4 & \end{array}$$

Here, $x - 1$ is divisor, $6x^2 - 7x + 5$ is divident, $6x - 1$ is quotient and 4 is remainder.

We know, dividend = divisor $\times$ quotient + remainder

Now, if we indicate the dividend by $f(x)$, the quotient by $h(x)$, the remainder by r and the divisor by $(x - a)$, from the above formula, we get,

$f(x) = (x - a) \cdot h(x) + r$, this formula is true to all values of a.

Substituting $x = a$ in both sides we get,

$f(a) = (a - a) \cdot h(a) + r = 0 \cdot h(a) + r = r$

Hence, $r = f(a)$

Therefore, if $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$. This formula is known as remainder theorem. That is, the remainder theorem gives the remainder when a polynomial $f(x)$ of positive degree is divided by $(x-a)$ without performing actual division. In the above example if $a = 1$, then $f(x) = 6x^2 - 7x + 5$.

$\therefore$ $f(1) = 6 - 7 + 5 = 4$ which is equal to the remainder. The degree of the divisor polynomial $(x-a)$ is 1, If the divisor is a factor of the dividend, the remainder will be zero and if it is not zero, the remainder will be a number other than zero.

Corollary 11. $(x - a)$ will be a factor of $f(x)$ if and only if $f(a) = 0$.

Proof: Let, $f(a) = 0$. Therefore, according to remainder theorem, if $f(x)$ is divided by $(x - a)$, the remainder will be zero. Hence, $(x - a)$ will be a factor of $f(x)$.

Conversely, let, $(x - a)$ is a factor of $f(x)$.

Therefore, $f(x) = (x - a) \cdot h(x)$, where $h(x)$ is a polynomial.

Putting $x = a$ in both sides we get,

$f(a) = (a - a) \cdot h(a) = 0$

$\therefore f(a) = 0$

Hence, any polynomial $f(x)$ will be divisible by $(x - a)$ if and only if $f(a) = 0$. This formula is known as factor theorem. □

Proposition 12. If degree of $f(x)$ is positive and $a \neq 0$, then if $f(x)$ is divided by $(ax + b)$, the remainder will be $f\left(-\frac{b}{a}\right)$.

Proof: Degree of divisor $ax + b$, $(a \neq 0)$ is 1.

Hence, we can write: $f(x) = (ax + b) \cdot h(x) + r = a\left(x + \frac{b}{a}\right) \cdot h(x) + r$

$\therefore f(x) = \left(x + \frac{b}{a}\right) \cdot a \cdot h(x) + r$

It is clear that, if $f(x)$ is divided by $\left(x + \frac{b}{a}\right)$, the quotient will be $a \cdot h(x)$ and the remainder will be r.

Here, divisor $= x - \left(-\frac{b}{a}\right)$

Hence, according to remainder theorem $r = f\left(-\frac{b}{a}\right)$

Therefore, if $f(x)$ is divided by $(ax + b)$, the remainder will be $\left(-\frac{b}{a}\right)$ □

Corollary 13. If $a \neq 0$, the expression $ax + b$ will be a factor of any polynomial $f(x)$, if and only if $f\left(-\frac{b}{a}\right) = 0$.

Proof: $a \neq 0$, $ax + b = a\left(x + \frac{b}{a}\right)$ will be a factor of $f(x)$, if and only if $\left(x + \frac{b}{a}\right)$ $= x - \left(-\frac{b}{a}\right)$ is a factor of $f(x)$. Hence, if and only if $f\left(-\frac{b}{a}\right) = 0$. This method of determining the factor using remainder theorem is called **Vanishing method.** □

Example 30. Resolve into factors: $x^3 - x - 6$

Solution: Here, $f(x) = x^3 - x - 6$ is a polynomial. The factors of the constant -6 are $\pm 1, \pm 2, \pm 3, \pm 6$.

Now, putting $x = 1, -1$ we see that the value of $f(x)$ is not zero.

But, putting $x = 2$ we see that the value of $f(x)$ is not zero.

Hence, $f(2) = 2^3 - 2 - 6 = 8 - 2 - 6 = 0$

Therefore, $x - 2$ is a factor of $f(x)$.

$\therefore f(x) = x^3 - x - 6$

$= x^3 - 2x^2 + 2x^2 - 4x + 3x - 6$

$= x^2(x - 2) + 2x(x - 2) + 3(x - 2)$

$= (x - 2)(x^2 + 2x + 3)$

Example 31. Resolve into factors: $x^3 - 3xy^2 + 2y^3 \quad x^2 + xy - 2y^2$.

Solution: Here, consider x a variable and y a constant.

We consider the given expression a polynomial of x.

Let, $f(x) = x^3 - 3xy^2 + 2y^3$

Then, $f(y) = y^3 - 3y \cdot y^2 + 2y^3 = 3y^3 - 3y^3 = 0$

$\therefore (x - y)$ is a factor of $f(x)$.

Now, $x^3 - 3xy^2 + 2y^3$

$= x^3 - x^2y + x^2y - xy^2 - 2xy^2 + 2y^3$

$= x^2(x - y) + xy(x - y) - 2y^2(x - y)$

$= (x - y)(x^2 + xy - 2y^2)$

Again let, $g(x) = x^2 + xy - 2y^2$

$\therefore g(y) = y^2 + y^2 - 2y^2 = 0$

$\therefore (x - y)$ is a factor of $g(x)$.

$\therefore g(x) = x^2 + xy - 2y^2$

$= x^2 - xy + 2xy - 2y^2$

$= x(x - y) + 2y(x - y)$

$= (x - y)(x + 2y)$

$\therefore x^3 - 3xy^2 + 2y^3 = (x - y)^2(x + 2y)$

Example 32. Resolve into factors: $54x^4 + 27x^3a - 16x - 8a$.

Solution: Let, $f(x) = 54x^4 + 27x^3a - 16x - 8a$

Then, $f\left(-\frac{1}{2}a\right) = 54\left(-\frac{1}{2}a\right)^4 + 27a\left(-\frac{1}{2}a\right)^3 - 16\left(-\frac{1}{2}a\right) - 8a$

$= \frac{27}{8}a^4 - \frac{27}{8}a^4 + 8a - 8a = 0$

$\therefore x - \left(-\frac{1}{2}a\right) = x + \frac{a}{2} = \frac{1}{2}(2x + a)$ is a factor of $f(x)$.

Hence, $(2x + a)$ is a factor of $f(x)$.

Now, $54x^4 + 27x^3a - 16x - 8a$

$= 27x^3(2x + a) - 8(2x + a)$

$= (2x + a)(27x^3 - 8)$

$= (2x + a)\{(3x)^3 - (2)^3\}$

$= (2x + a)(3x - 2)(9x^2 + 6x + 4)$

Example 33. $g(a) = a^3 + a^2 + 10a - 8$, $f(a) = a^3 - 9 + (a + 1)^3$.

1) If $g(a)$ is divided by $(a - 2)$ then determine the remainder
2) Resolve into factors: $f(a)$

Solution: 1) Given, $g(a) = a^3 + a^2 + 10a - 8$

according to remainder theorem, if $g(a)$ is divided by $(a - 2)$ then the remainder will be $g(2)$.

$\therefore\ g(2) = 2^3 + 2^2 + 10 \cdot 2 - 8 = 8 + 4 + 20 - 8 = 32 - 8 = 24$

$\therefore\ g(2) = 24$

Calculated remainder is 24.

$f(a) = a^3 - 9 + (a + 1)^3$

$f(a)$ is a polynomial, if we put $a = 1$, then the result of the polynomial will be zero.

Therefore $(a - 1)$ is a factor of the polynomial.

$\therefore\ f(a) = a^3 - 9 + a^3 + 3a^2 + 3a + 1 = 2a^3 + 3a^2 + 3a - 8$

$= 2a^3 - 2a^2 + 5a^2 - 5a + 8a - 8$

$= 2a^2(a - 1) + 5a(a - 1) + 8(a - 1)$

$= (a-1)(2a^2+5a+8)$

$\therefore\ a^3-9+(a+1)^3 = (a-1)(2a^2+5a+8)$

Work: Resolve into factors:

1) $x^3-21x-20$ 2) $2x^3-3x^2+3x-1$ 3) $x^3+6x^2+11x+6$

Exercises 3.4

Resolve into factors:

1. $3a^3+2a+5$
2. $x^3-7xy^2-6y^3$
3. x^3+2x^2-5x-6
4. x^3+4x^2+x-6
5. $a^3+3a+36$
6. a^4-4a+3
7. $a^3-a^2-10a-8$
8. x^3-3x^2+4x-4
9. $a^3-7a^2b+7ab^2-b^3$
10. x^3-x-24
11. $x^3+6x^2+11x+6$
12. $2x^4-3x^3-3x-2$
13. $4x^4+12x^3+7x^2-3x-2$
14. $x^6-x^5+x^4-x^3+x^2-x$
15. $4x^3-5x^2+5x-1$
16. $18x^3+15x^2-x-2$

Forming and applying algebraic formulae in solving real life problems

In our daily business we face the realistic problems in different time and in different ways. These problems are described linguistically. In this section, we shall discuss the formation of algebraic formulae and their applications in solving different problems of real surroundings which are described linguistically. As a result of this discussion the students on the one hand, will get the conception about the application of mathematics in real surroundings, on the other hand, they will be eager to learn mathematics for their understanding of the involvement of mathematics with their surroundings.

Methods of solving the problems:

1. At first the problem will have to be observed carefully and to read attentively and then to identify which are unknown and which are to be determined.

2. One of the unknown quantities is to be denoted with any variable (say x). Then realising the problem well, express other unknown quantities in terms of the same variable x.
3. The problem will have to be splitted into small parts and express them by algebraic expressions.
4. Using the given conditions, the small parts together are to be expressed by an equation
5. The value of the unknown quantity x is to be found by solving the equation.

Different formulae are used in solving the problems based on real life. The formulae are mentioned below :

Related to Payable or Attainable

Suppose, q = amount of money payable or attainable per person

n = number of person

$\therefore$ Amount of payable or attainable money, $A = qn$

Related to Time and Work

Suppose, q = portion of a work performed by every one in unit of time

n = number of performing of work

x = total time of doing work

W = portion of a work done by n persons in time x

$\therefore W = qnx$

Related to Time and Distance

Suppose, v = speed per hour

t = total time

d = total distance

$\therefore d = vt$

Related to pipe and water tank

Suppose, Q_0 = amount of stored water in a tank at the time of opening the pipe

q = amount of water flowing in or flowing out by the pipe in a unit time.

t = time taken.

$Q(t)$ = amount of water in the tank in time t

$\therefore\ Q(t) = Q_0 \pm qt$

'+' sign at the time of flowing of water in and '−' sign, at the time of flowing of water out are to be used.

Related to percentage

Suppose, b = total quantity

r = rate of percentage $= \frac{s}{100} = s\%$

p = part of percentage $s\%$ of $= b$

$\therefore\ p = br$

Related to profit and loss

Suppose, C = cost price

r = rate of profit or loss

$\therefore$ selling price $S = C(1 \pm r)$

in case of profit, $S = C(1 + r)$

in case of loss, $S = C(1 - r)$

Related to investment and profit

Suppose, I = profit after time n

n = specific time

P = principal

r = profit of unit principal at unit time

A = principal with profit after time n

In the case of simple profit,

$I = Pnr$

$A = P + I = P + Pnr = P(1 + nr)$

In the case of compound profit,

$A = P(1 + r)^n$

Example 34. For a function of Annual Sports, members of an association made a budget of Tk. 45000 and divided that every member would subscribe equally. But 5 members refused to subscribe. As a result, amount of subscription of each member increased by Tk. 15 per head. How many members were in the association?

Solution: Let the number of members of the association be x and amount of subscription per head be Tk. q.

Then total amount of subscription, Tk $A = qx = 45,000$.

Actually numbers of members were $(x - 5)$ and amount of subscription per head became Tk. $(q + 15)$.

Then, total amount of subscription $(x - 5)(q + 15)$

By the question,

$$qx = (x - 5)(q + 15) \cdots\cdots (1)$$

$$qx = 45000 \cdots\cdots (2)$$

From equation (1) we get,

$$qx = (x - 5)(q + 15)$$

or, $qx = qx - 5q + 15x - 75$

or, $5q = 15x - 75 = 5(3x - 15)$

$\therefore q = 3x - 15$

Putting the value of q in (2),

$$(3x - 15) \times x = 45000$$

or, $3x^2 - 15x = 45000$

or, $x^2 - 5x = 15000$ [dividing both sides by 3]

or, $x^2 - 5x - 15000 = 0$

or, $x^2 - 125x + 120x - 15000 = 0$

or, $x(x - 125) + 120(x - 125) = 0$

or, $(x - 125)(x + 120) = 0$

Therefore, $(x - 125) = 0$ or $(x + 120) = 0$

or, $x = 125$ or, $x = -120$

Since the number of members cannot be negative, hence the value of x as -120 is not acceptable.

Therefore, number of members of the association is 125

Example 35. Rafiq can do a work in 10 days and Shafiq can do that work in 15 days. In how many days do they together finish the work?

Solution: Let, they can finish the work in d days.

Name	Days to complete	Part done in 1 day	Part done in d day
Rafiq	10	$\frac{1}{10}$	$\frac{d}{10}$
Shafiq	15	$\frac{1}{15}$	$\frac{d}{15}$

As per question, $\frac{d}{10} + \frac{d}{15} = 1$ or, $d\left(\frac{1}{10} + \frac{1}{15}\right) = 1$

or, $d\left(\frac{3+2}{30}\right) = 1$ or, $\frac{5d}{30} = 1$

or, $d = \frac{30}{5} = 6$

Therefore, they together can finish the work in 6 days.

Example 36. A boatman can row x km in time t_1 hour against the current. To cover that distance along the current he takes t_2 hour. How much is the speed of the boat and the current?

Solution: Let the speed of the boat in still water be u km per hour and that of the current be v km per hour.

Then, along the current, the effective speed of boat is $(u + v)$ km per hour and against the current, the effective speed of boat is $(u - v)$ km per hour.

We know, speed $= \frac{\text{distance traversed}}{\text{time}}$

According to the question, $u + v = \frac{x}{t_2} \cdots\cdots (1)$

and $u - v = \frac{x}{t_1} \cdots\cdots (2)$

Adding equations (1) and (2) we get,

$2u = \frac{x}{t_2} + \frac{x}{t_1} = x\left(\frac{1}{t_1} + \frac{1}{t_2}\right)$ or, $u = \frac{x}{2}\left(\frac{1}{t_1} + \frac{1}{t_2}\right)$

Subtracting equation (2) from equation (1) we get,

$2v = \frac{x}{t_2} - \frac{x}{t_1} = x\left(\frac{1}{t_2} - \frac{1}{t_1}\right)$ or, $v = \frac{x}{2}\left(\frac{1}{t_2} - \frac{1}{t_1}\right)$

Hence speed of current is $\frac{x}{2}\left(\frac{1}{t_2} - \frac{1}{t_1}\right)$ km per hour and speed of boat is $\frac{x}{2}\left(\frac{1}{t_1} + \frac{1}{t_2}\right)$ km per hour.

Example 37. A pipe can fill up an empty tank in 12 minutes. Another pipe flows out 14 litre of water per minute. If the two pipes are opened together and the empty tank is filled up in 96 minutes, how much water does the tank contain ?

Solution: Let x litre of water flows in per minute by the first pipe and the tank can contain y litre of water.

According to the question, the tank is filled up by first pipe in 12 minutes

$\therefore y = 12x \cdots\cdots (1)$

Again, the empty tank is filled up by the two pipes together in 96 minutes.

$\therefore y = 96x - 96 \times 14 \cdots\cdots (2)$

From equation (1) we get, $x = \frac{y}{12}$

putting the value of x in equation (2) we get,

$y = 96 \times \frac{y}{12} - 96 \times 14$

or, $y = 8y - 96 \times 14$

or, $7y = 96 \times 14$

or, $y = \frac{96 \times 14}{7} = 192$

Hence, total 192 litre of water is contained in the tank.

Work:

1) For a picnic, a bus was hired at Tk. 2400 and it was decided that every passenger would have to give equal fare. But due to the absence of 10 passengers, fare per head was increased by Tk. 8. How many passengers did go by the bus and how much money did each of the passengers give as fare?
2) A and B together can do a work in p days. A alone can do that work in q days. In how many days can B alone do the work ?
3) A person rowing against the current can go 2 km per hour. If the speed of the current is 3 km per hour, how much time will he take to cover 32 km, rowing along the current ?

Example 38. Price of a book is Tk. 24. This price is 80% of the actual price. The Government subsidize the due price. How much money does the Government subsidize for each book?

Solution: Market price = 80% of the actual price

We know, $p = br$

Here, $p = 24$ Tk and $r = 80\% = \dfrac{80}{100}$

$\therefore 24 = b \times \dfrac{80}{100}$

or, $b = \dfrac{24 \times 100}{80}$

$\therefore b = 30$ Tk.

Hence, the actual price of the book is Tk. 30.

$\therefore$ Amount of subsidized money Tk. $= (30 - 24)$ or Tk. $= 6$

Hence, subsidized money for each book is Tk. 6.

Example 39. The loss is $r\%$ when n oranges are sold per taka. How many oranges are to be sold per taka to make a profit of $s\%$?

Solution: If the cost price is Tk 100, the selling price at the loss of $r\%$ is Tk $(100 - r)$.

If selling price is Tk. $(100 - r)$, cost price is Tk. 100

$\therefore$ If selling price is Tk. 1, cost price is Tk. $\frac{100}{100-r}$.

Again, if cost price is Tk. 100, selling price at the profit of $s\%$ is Tk. $(100+s)$.

$\therefore$ if cost price is Tk. $\frac{100}{100-r}$, selling price at the profit of $s\%$ is Tk. $\left(\frac{100+s}{100} \times \frac{100}{100-r}\right)$

$= \text{Tk. } \frac{100+s}{100-r}$

Therefore, in Tk. $\frac{100+s}{100-r}$ number of oranges is to be sold is n.

$\therefore$ $InTk.1$ number of oranges is to be sold is $n \times \left(\frac{100-r}{100+s}\right)$

Hence, $\frac{n(100-r)}{100+s}$ oranges are to be sold per taka.

Example 40. What is the profit of Tk. 650 in 6 years at the rate of profit Tk. 7 percent per annum ?

Solution: We know, $I = Pnr$

Here, $P = 650$ Tk, $n = 6$ year, percent per annum $s = 7$ Tk

$\therefore r = \frac{s}{100} = \frac{7}{100}$

$\therefore I = 650 \times 6 \times \frac{7}{100} = 273$

Hence, profit is Tk. 273.

Example 41. Find the compound principal and compound profit of Tk. 15000 in 3 years at the profit of 6 percent per annum.

Solution: We know, $C = P(1+r)^n$ [where C is the profit principal in the case of compound profit]

Given, $P = 15000$ Tk, $r = 6\% = \frac{6}{100}$, $n = 3$ year

$\therefore C = 15000\left(1+\frac{6}{100}\right)^3 = 15000\left(1+\frac{3}{50}\right)^3 = 15000\left(\frac{53}{50}\right)^3$

$$= 15000 \times \frac{53}{50} \times \frac{53}{50} \times \frac{53}{50} = \frac{446631}{25} = 17865.24$$

$\therefore$ Compound principal is Tk 17865.24

$\therefore$ Compound profit is Tk. $(17865.24 - 15000)$ Tk. $= 2865.24$.

Work:

1) The loss is 50% when 10 lemons are sold per taka. What will be the profit if 6 lemons are sold per taka?
2) What will be the profit of principal of Tk. 750 in 4 years at simple interest $6\frac{1}{2}$ percent per annum?
3) Find the compound interest of Tk. 2000 in 3 years at compound interest of Tk. 4 percent per annum.

Example 42. The loss is $x\%$ when 10 ice creams are sold per taka. How many ice creams are to be sold per taka to make the profit of $z\%$?

Solution: If the cost price is Tk. 100, the selling price at the loss of $x\%$ is Tk. $= (100 - x)$

If selling price is Tk. $(100 - x)$ cost price is Tk. 100

$\therefore$ If selling price is Tk. 1 cost price is Tk. $\dfrac{100}{100 - x}$

Hence, the cost price of 10 ice creams is Tk. $\dfrac{100}{100 - x}$

$\therefore$ The cost price of 1 ice cream is Tk. $\dfrac{100}{(100 - x) \times 10}$.

Again if the cost price is Tk. 100 selling price at the profit of $z\%$ is Tk. $(100 + z)$

If the cost price is Tk. 100 selling price is Tk. $(100 + z)$.

If the cost price is Tk. 1 selling price is Tk. $\dfrac{100 + z}{100}$

$\therefore$ If the cost price is Tk. $\dfrac{100}{(100 - x) \times 10}$

selling price is Tk. $\dfrac{100 + z}{100} \times \dfrac{100}{(100 - x) \times 10} = \dfrac{(100 + z)}{(100 - x) \times 10}$

The selling price of 1 ice cream is Tk. $\dfrac{(100 + z)}{(100 - x) \times 10} = \dfrac{100 + z}{1000 - 10x}$

Hence $\frac{1000 - 10x}{100 + z}$ ice creams have to be sold per taka.

Exercises 3.5

1. If $f(x) = x^2 - 4x + 4$, which one of the following is the value of $f(2)$?
   1) 4 2) 2 3) 1 4) 0
2. Which one of the following is the value of $\frac{1}{2}\{(a+b)^2 - (a-b)^2\}$?
   1) $2(a^2 + b^2)$ 2) $a^2 + b^2$ 3) $2ab$ 4) $4ab$
3. If $x + \frac{2}{x} = 3$, what is the value of $x^3 + \frac{8}{x^3}$?
   1) 1 2) 8 3) 9 4) 16
4. Which one of the following is the factorized form of $p^4 + p^2 + 1$?
   1) $(p^2 - p + 1)(p^2 + p - 1)$ 2) $(p^2 - p - 1)(p^2 + p + 1)$
   3) $(p^2 + p + 1)(p^2 + p + 1)$ 4) $(p^2 + p + 1)(p^2 - p + 1)$
5. If $x = 2 - \sqrt{3}$, then what is the value of x^2?
   1) 1 2) $7 - 4\sqrt{3}$
   3) $2 + \sqrt{3}$ 4) $\frac{1}{2 - \sqrt{3}}$
6. If $f(x) = x^2 - 5x + 6$ and $f(x) = 0$, what is the value of $x =$?
   1) $2, 3$ 2) $-5, 1$ 3) $-2, 3$ 4) $1, -5$
7. What is to be added to $9x^2 + 16y^2$ so that their sum will be a perfect square?
   1) $6xy$ 2) $12xy$ 3) $24xy$ 4) $144xy$

   If $x^4 - x^2 + 1 = 0$, answer the following questions from 8 to 10

8. What is the value of $x^2 + \frac{1}{x^2}$?
   1) 4 2) 2 3) 1 4) 0
9. What is the value of $(x + \frac{1}{x})^2$?
   1) 4 2) 3 3) 2 4) 0
10. What is the value of $x^3 + \frac{1}{x^3}$?
    1) 3 2) 2 3) 1 4) 0
11. If $a^2 + b^2 = 9$ and $ab = 3$

    (*i*) $(a - b)^2 = 3$

(*ii*) $(a+b)^2 = 15$

(*iii*) $a^2 + b^2 + a^2b^2 = 18$

Which one of the following is correct?

1) *i*, *ii* 2) *i*, *iii* 3) *ii*, *iii* 4) *i*, *ii* and *iii*

12. If $3a^5 - 6a^4 + 3a + 14$ is a algebraic expression-

(*i*) variable of the expression a

(*ii*) degree of the expression 5

(*iii*) constant of a^4 is 6

Which one of the following is correct?

1) *i*, *ii* 2) *i*, *iii* 3) *ii*, *iii* 4) *i*, *ii* and *iii*

13. Factor of $p^3 - \frac{1}{64}$ -

(*i*) $p - \frac{1}{4}$

(*ii*) $p^2 + \frac{p}{4} + \frac{1}{8}$

(*iii*) $p^2 + \frac{p}{4} + \frac{1}{16}$

Which one of the following is correct?

1) *i*, *ii* 2) *i*, *iii*

3) *ii*, *iii* 4) *i*, *ii* and *iii*

14. A can do a work in p days and B can do it in $2p$ days. They started to do the work together and after some days A left the work unfinished. B completed the rest of the work in r days. In how many days was the work finished ?

15. 10 persons can do a work in 7 days by working 6 hours a day. Working how many hours per day can 14 persons finish the work in 6 days?

16. Mita can do a work in 10 days. Rita can do that work in 15 days.In how many days will they together complete the work ?

17. A bus was hired at Tk. 5700 to go for a picnic under the condition that every passenger would bare equal fare. But due to the absence of 5 passengers, the fare was increased by Tk. 3 per head. How many passengers availed the bus ?

18. A boatman can go d km in p hours against the current. He takes q hours to cover that distance along the current. What is the speed of the current and the boat ?

19. A boatman goes 15 km and returns from there in 4 hours plying by oar. He goes 5 km at a period of time along the current and goes 3 km at the same period of time against the current. Find the speed of the oar and current.

20. Two pipes are connected with a tank. The empty tank is filled up in t_1 minutes by the first pipe and it becomes empty in $t_2$minutes by the second pipe. If the two pipes are opened together, in how much time will the tank be filled up ?(Here $t_2 > t_1$)

21. A tank is filled up in 12 minutes by a pipe. Another pipe flows out 15 litre of water in 1 minute. When the tank remains empty, the two pipes are opened togethr and the tank is filled up in 48 minutes. How much water does the tank contain ?

22. Divide Tk. 260 among A, B and C in such a way that 2 times the share of A, 3 times the share of B and 4 times the share of C are equal to one another.

23. Due to the selling of a commodity at the loss $x\%$ such price is obtained that due to the selling at the profit of $3x\%$ Tk. $18x$ more is obtained. What was the cost price of the commodity ?

24. If a pen is sold at Tk. 11, there is a profit of 10%. What was the cost price of the pen ?

25. Due to the sale of a notebook at Tk. 36, there was a loss. If the notebook would be sold at Tk. 72, there would be profit amounting twice the loss. What was the cost price of the notebook ?

26. If the simple profit of Tk. 300 in 4 years and the simple profit of Tk. 400 in 5 years together are Tk. 148, what is the percentage of profit ?

27. If the difference of simple profit and compound profit of some principal in 2 years is Tk. 1 at the rate of profit 4%, what is the principal ?

28. Some principal beomes Tk. 460 with simple profit in 3 years and Tk. 600 with simple profit in 5 years. What is the rate of profit ?

29. How much money will become Tk. 985 as profit principal in 13 years at the rate of simple profit 5% per annum ?

30. How much money will become Tk. 1248 as profit principal in 12 years at the rate of profit 5% per annum ?

31. Find the difference of simple profit and compound profit of Tk. 8000 in 3 years at the rate of profit 5%.

32. The Value Added Tax (VAT) of sweets is $x\%$. If a trader sells sweets at Tk. P including VAT, how much VAT is he to pay ? If $x = 15$, $P = 2300$, what is the amount of VAT ?

33. Sum of a number and its multiplicative inverse is 3.

    1) Taking the number as the variable x, express the above information by an equation

    2) Find the value of $x^3 - \frac{1}{x^3}$.

    3) Prove that, $x^5 + \frac{1}{x^5} = 123$

34. Each of the members of an association decided to subscribe 100 times the number of members. But 7 members did not subscribe. As a result, amount of subscription for each member was increased by Tk. 500 than the previous.

    1) If the number of members is x and total amount of subscription is Tk. A, find the relation between them.

    2) Find the number of members of the association and total amount of subscription.

    3) $\frac{1}{4}$ of total amount of subscription at the rate of simple profit 5% and rest of the money at the rate of simple profit 4% were invested for 2 years. Find the total profit.

35. A bus was hired at Tk. 2400 to go for a picnic under the condition that every passenger would bare equal fare. But due to the absence of 10 passengers, the fare was increased by Tk. 8 per head.

    1) Determine the increased fare per head and percentage of absent passengers.

    2) Determine the fare per head of the passengers.

    3) Find the difference of simple profit and compound profit of the amount equivalent to the bus fare in 13 years at the rate of profit 5%.

36. Going from point A to point B in a canal, it has to be returned. If the speed of oar is constant then which time will be greater if there is current or not?

37. The grass in a field increases at a constant rate. 17 cows can eat up all grass in 30 days whereas it takes 24 days for 19 cows. 4 cows were sold from a herd after the herd had eaten grass for 6 days. It took 2 more days for the herd to eat up the grass. How many cows were in that herd?

38. Two brothers had a trained horse which can follow any order. Getting out from the house at the same time two brothers wanted to go to a Baishakhi fair which is 20 miles away from their house. The hourse can carry only one brother at any moment. If the speed of each brother is 4 miles per hour and speed of the horse(with or without person) is 10 miles per hour, then what is the minimum time they need to reach the fair? How much distance each brother has to walk?